

**Show Me the Money: How Foreign Direct Investment Influences Trade in
ASEAN countries**

Leo Mai

Denison University

Data Analytics 401 - Senior Seminar

Dr. Alexandre Scarcioffolo

Spring 2025

Abstract

This paper investigates how foreign direct investment (FDI) affects international trade flows in **ten member countries of the Association of Southeast Asian Nations (ASEAN)**, using data from 2010 to 2023. Using fixed effects models, the study examines the relationship between inward FDI and both export and import values, controlling for **Gross Domestic Product (GDP)**, unemployment rate, population, **industrial level**, and **economic freedom**. Initial baseline models reveal no significant effect of FDI on trade, suggesting heterogeneous outcomes across countries. To account for this, the study introduces GDP as an interaction term with FDI, revealing that inward FDI is more trade-enhancing in lower GDP countries. This finding supports theories of less developed economies benefiting more from FDI, often through mechanisms such as technology transfer and productivity gains. These results contribute to the literature by highlighting the conditional nature of the relationship between FDI and trade, especially in heterogeneous regions such as ASEAN.

1. Introduction

The Association of Southeast Asian Nations (ASEAN) consists of ten out of eleven countries in the Southeast Asia region: Brunei Darussalam, Myanmar, Cambodia, Indonesia, Laos, Malaysia, the Philippines, Singapore, Thailand and Vietnam. Over the past 50 years, the region has experienced rapid economic growth, driven initially by Singapore, Thailand, and Indonesia, and more recently by Vietnam, Myanmar, and Laos (Intal, Chen, et al., 2017). Furthermore, ASEAN is a top destination for foreign direct investment (FDI), maintaining the lead for three consecutive years as of 2024 (The ASEAN Secretariat & United Nations Trade and Development, 2024). Given the importance of ASEAN countries in the global supply chain as manufacturing powerhouses and the importance of FDI in ASEAN economies, understanding the relationship between FDI and trade activities is crucial for the development of the region.

Existing literature provides evidence of mixed quality for this question. Current research is narrow in scope, either restricted to certain sectors or certain countries within ASEAN. This paper fills in this gap in knowledge by providing a comprehensive, cross-country, sector-agnostic analysis of the relationship between FDI and trade activity of all ten ASEAN member countries.

Inward FDI, or FDI inflows, are investments made by businesses from foreign countries into an enterprise of the host country. Given this definition, this paper investigates how inward FDI affects international trade flows in ASEAN countries from 2010 to 2023. Existing theories assume that FDI stimulates economic development of the host country, which can lead to increased trade activity. Using fixed effects regression models and controlling for key macroeconomic factors, this paper examines both the overall effect of FDI and whether this effect varies based on the GDP level of a country. The findings show no significant relationship between FDI and trade in aggregate, but reveal a positive and significant effect of FDI on trade in countries with lower GDP, suggesting that less developed ASEAN economies may benefit more from FDI inflows. Although there may

be other social, political, or institutional factors that may also impact trade activity of a country, this study will focus on socioeconomic factors that are believed to be most relevant to trade activities. Nevertheless, it contributes to the literature by providing empirical evidence on the conditional nature of the impact of FDI on trade in a heterogeneous regional bloc like ASEAN.

The rest of this paper proceeds as follows: **Section 2** reviews the current literature surrounding FDI and trade; **Section 3** describes the data source, variables, and any pre-processing steps required prior to analysis; **Section 4** defines the choice of methodology, the underlying mathematical theory and the justification of such choice; **Section 5** presents the model results; **Section 6** discusses the interpretation of such results and limitations; **Section 7** raises ethical considerations and implications; and **Section 8** draws final conclusions.

2. Literature Review

FDI is found by existing literature to be mutually beneficial to both the businesses and the hosting country, in terms of economic well-being. For the investing foreign businesses, Dunning (2001) describes three motivations for a business to utilize FDI using his OLI paradigm: ownership advantages (O) - knowledge and privileges that the firm possesses that are not available to domestic competitors, location advantages (L) - such as better infrastructure or cheap production inputs, and internalization advantages (I) - assets that the firm can better use itself, rather than selling or leasing to other firms. From Dunning's theory, Wadhwa and Reddy (2011) identified four types of FDI based on their purpose: market-seeking FDI that helps businesses penetrate the local and regional markets; resource-asset seeking FDI that enables access to raw natural resources, cheap labor pools, physical infrastructure and technology level; efficiency-seeking FDI that seeks to minimize costs of production; and strategic asset-seeking FDI that integrates the firm's regional and global strategy into strategic foreign networks of assets like technology, organizational abilities and markets (Faeth, 2009). For the host economy, FDI can be an

engine that drives development and growth, enabling technology transfer, increasing employment, and allowing access to high-quality managerial and employee training ((Borensztein, De Gregorio, & Lee, 1998; Zhang, 2001; Alfaro, 2016). However, FDI can also weaken domestic firms, society and government control (Zhang, 2001), or the host country may lack sufficient human capital to enable the increased productivity that FDI provides (Borensztein et al., 1998).

Besides looking at the overall economy, academic literature also looks at the effects of FDI on international trade specifically. Through this lens, there are two FDI patterns: firms replicating their existing production in foreign markets (horizontal FDI), and firms investing in different places on the supply chain (vertical FDI). Thus, horizontal FDI tends to substitute trade (as goods can now be created locally instead of imported), while vertical FDI tends to stimulate trade (usually to the home country of the foreign firm) (Aizenman & Noy, 2006). However, the direction of causality between FDI level and trade activity is debated, with some claiming that higher FDI leads to an increase in trade activity (Barry & Bradley, 1997; Africano, Magalhães, et al., 2005), some others claiming vice versa (Ghosh (2007), and a few claiming a bidirectional relationship between the two (Liu, Wang, & Wei, 2001; Aizenman & Noy, 2006). This is understandable, as there are a lot of real-life interdependency between macroeconomic indicators like FDI and trade activity, among others like GDP and inflation.

Another area of interest for this paper in academic literature is the study of FDI and trade that is specific to the ASEAN region. This region is an interesting one for research because it is one of the fastest growing regions in the world, with a mix of middle- and high-income economies (Intal et al., 2017). With many developing economies in the region such as Indonesia, the Philippines and Vietnam, the **ASEAN** region is an attractive destination for FDI by developed economies (Magasházi, 2015). Furthermore, ASEAN, as an integrated market, has established free trade agreements with many of its surrounding economies, including China, South Korea, Japan, Australia, and New Zealand (Okabe et

al., 2015), and ASEAN itself is a free trade area. On this topic, Magasházi (2015) found that trade from ASEAN to other countries, as well as its inward FDI flows, increased after the establishment of the aforementioned free trade agreements (FTAs). And while Magasházi focused on ASEAN in the global context, Li, Scollay, and Maani (2016) and Li and Maani (2018) focused on the relationship between the region and China, and found that the ASEAN - China Free Trade Area (ACFTA) attracted both horizontal and vertical FDI.

This study fills the gap in existing literature in several ways. First, current papers mostly only focus on the relationship between trade and FDI in the manufacturing sector. Second, there are limited research into the relationship between FDI and trade activity among ASEAN economies, rather than between FDI and FTA establishments. Third, research concerning the region focuses on only a subset of ASEAN countries, most commonly ASEAN-5 or ASEAN-6 members - Indonesia, Malaysia, Philippines, Singapore, Thailand (ASEAN-5) and Vietnam (ASEAN-6). This study examines the effect of FDI across all sectors on trade value of all ten ASEAN member economies, using a fixed effects model.

3. Data

The data for this study contains annual statistics for the ten ASEAN member countries, from 2010 to 2023. The statistics include net FDI inflow value, import and export values, gross domestic product (GDP) value, unemployment rate, population, industry (as percentage of GDP), and the Heritage Foundation's Index of Economic Freedom (IEF). These variables are chosen because they directly relate to the research question of how FDI affects trade in ASEAN countries, and the time frame is selected based on the availability of data. **Table 1** summarizes the variables used in this study.

In terms of data source, the FDI inflow, import and export values were collected from ASEANstats, ASEAN's official data portal, the IEF was collected from the Heritage Foundation's website, and the remaining statistics were collected from the World Bank.

Table 1
Variables Summary

Variable	Source	Unit	Description
Inward FDI	ASEANstats	USD millions	Annual FDI inflows to each ASEAN country
Exports	ASEANstats	USD millions	Total value of goods and services exported by each country
Imports	ASEANstats	USD millions	Total value of goods and services imported by each country
GDP	World Bank	USD millions	Gross Domestic Product by each country
Unemployment Rate	World Bank	Percentage	Percentage of the labor force that is unemployed but actively seeking work
Population	World Bank	Number of persons	Total population of each country
Industry	World Bank	Percentage of GDP	Proportion of GDP that comes from the industrial sector
IEF	Heritage	NA	Measure the degree of economic freedom in the world's nations

Since these data sources are data reported by the nations themselves (ASEANstats), or collected by a credible international financial institution (World Bank), the reliability, consistency, and accuracy of the dataset is ensured.

In terms of licensing, ASEANstats' Terms and Conditions of Use allows the statistical information by ASEANstats to be used, downloaded and copied, provided that ASEANstats is duly credited and acknowledged as the data source. As such, citing them is sufficient for me to use the data available on the website for analysis. Data from the World Bank, whose author is the World Bank Group, is licensed under CC BY-4.0, allowing the use of data when given proper credit, similar to ASEANstats' Terms and Conditions of use. Finally, the Heritage Foundation does not explicitly license its Index. However, since the data is publicly and freely available, I will follow common practices of using publicly

available data to handle this data source, which includes properly crediting the Heritage Foundation as the data owner.

The multiple data sources were aggregated into a single dataset. Each query to either data source yielded one variable, with the exception of export and import value, which were simultaneously collected in one query. The data was then pivoted in Excel such that each observation is for one country, for one year, and then the variables are put into a single file. Finally, the FDI inflow, import, export, and GDP variables were converted from USD to millions USD using R. The data files and code used are available in **Appendix B**.

4. Methods

Given the panel structure of the data, coupled with the heterogeneity nature of ASEAN nations (Intal et al., 2017), this study uses fixed effects models to examine the relationship between trade value and FDI. The model performs regression analysis on the variables specified above, while controlling for country and year fixed effects. Two batches of models are created: one baseline model, and one model where GDP is introduced as an interaction term with FDI. As such, the specification for the baseline model is as follows:

$$\begin{aligned} \text{Trade}_{it} = & \beta_C + \beta_Y + \beta_1 \cdot \text{FDI}_{it} + \beta_2 \cdot \text{GDP}_{it} + \beta_3 \cdot \text{UnempRate}_{it} \\ & + \beta_4 \cdot \text{Population}_{it} + \beta_5 \cdot \text{Industry}_{it} + \beta_6 \cdot \text{EconFreedom}_{it} + \varepsilon_{it} \end{aligned} \quad (1)$$

And the formula for our model with the interaction term is as follows:

$$\begin{aligned} \text{Trade}_{it} = & \beta_C + \beta_Y + \beta_1 \cdot \text{FDI}_{it} + \beta_2 \cdot \text{GDP}_{it} + \beta_3 \cdot (\text{FDI}_{it} \times \text{GDP}_{it}) \\ & + \beta_4 \cdot \text{UnempRate}_{it} + \beta_5 \cdot \text{Population}_{it} \\ & + \beta_6 \cdot \text{Industry}_{it} + \beta_7 \cdot \text{EconFreedom}_{it} + \varepsilon_{it} \end{aligned} \quad (2)$$

Where:

- Trade_{it} is either exports or imports for country i at time t
- FDI_{it} is the FDI inflow for country i at time t
- GDP_{it} is the GDP value for country i at time t
- UnempRate_{it} is the unemployment rate for country i at time t
- Population_{it} is the population for country i at time t
- **Industry_{it} is the industry level** for country i at time t
- **EconFreedom_{it} is the IEF** for country i at time t
- β_C is the country fixed effects
- β_Y is the year fixed effects
- ε_{it} is the error term

To further investigate the robustness of the models and capture different levels of unobserved heterogeneity, each model is estimated under two fixed effects specifications. The first includes only country fixed effects, while the second includes both country and year fixed effects. This dual specification helps distinguish the influence of country-specific characteristics from year-specific shocks, such as global financial events or policy changes that may affect all countries in a given year. Furthermore, the different types of FDI - horizontal or vertical, or the four types specified by Wadhwa and Reddy - can influence the two dimensions of trade differently, so it is necessary to examine imports and exports separately.

The model design is justified for several reasons. First, the diversity of ASEAN member economies mean that there are time-invariant characteristics of each country that can affect both FDI attractiveness and trade activity. For example, Singapore's geographic location and consistent pro-foreign investment, export-oriented government policies since

1965 leads to its high FDI inflows and trade value (Pangarkar & Lim, 2003; Feridun & Sissoko, 2011). As such, the country fixed effect is introduced to control for these characteristics, allowing the model to capture variability within each ASEAN country. Second, considering that the time frame of our data contains economic shocks like the 2008 financial crisis and the COVID-19 pandemic - which could simultaneously affect all countries, the year fixed effect is also introduced. Third, the introduction of GDP as an interaction term is reasonable, as FDI amount can be conditional on GDP level. For example, using Wadhwa and Reddy's FDI classification, resource-seeking FDI may prefer country with low GDP for cheap labor, or high GDP for better infrastructure. Fourth, fixed effects models has been used in empirical analysis of panel data. For example, among the mentioned literature in this paper, Aizenman and Noy (2006) and Li et al. (2016) incorporated fixed effect models on their research into FDI and trade. **However, it should be noted that although fixed effects regression reduces certain biases by controlling for unobserved, country-specific characteristics, it does not establish causality. This paper's findings reflect conditional correlations between FDI and trade, rather than definitive causal impacts.** To sum up, the data structure, ASEAN countries' heterogeneity, and historical usage of the fixed effects methodology justify the design of this model, whose result is shown in the following section.

5. Results

The first batch of models estimate the impact of FDI inflows, unemployment rate, population, **industrial level and economic freedom** on import and export values, respectively. For each trade statistic, two fixed effects specifications are used: country fixed effects only, and both country and year fixed effects. This creates four total models, whose results for FDI coefficients and overall model statistics are shown in **Table 2**. The full regression results of all variables are reported in **Appendix Table A1**.

For this first group of models, the effect of FDI inflows is positive for both import value models and negative for both export models. However, these FDI inflows coefficients

Table 2*Baseline Model Group*

Variable	Export Model 1	Export Model 2	Import Model 1	Import Model 2
FDI Inflows	-0.0118 (0.2998)	-0.0540 (0.2511)	0.2781 (0.3431)	0.2428 (0.2826)
Country Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	No	Yes	No	Yes
R ²	0.9661	0.9701	0.9526	0.9599
Within R ²	0.6017	0.3929	0.5012	0.2398
Observations	136	136	136	136

Standard errors in parentheses. Full regression results are reported in Appendix Table A1.

Significance codes: . p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

are statistically significant across all four models. As for the controlling variables, both GDP and unemployment rate has a positive and statistically significant effect on both import and export. While population has a negative effect on both trade values, but it is only statistically significant for the export models. **The effect of industry level is positive, but only statistically significant in the model with just country fixed effects. Finally, economic freedom as measured by the IEF also has a positive effect, but they are not statistically significant.** Across all four models, the R-squared values are all at around 96%, and the Within R-squared values range between 24% and 60%. The models that include both country and year fixed effects consistently have lower Within R-squared values than their counterparts with only country fixed effects.

The second batch of models is similar to the first batch, with the addition of GDP as an interaction term to FDI inflows. The results for FDI inflows and the interaction term are shown in **Table 3**. The full regression results of all variables are reported in **Appendix Table A2**.

The results for the second batch of models are more consistent than that of the first batch. The coefficients for inward FDI are all positive, while the coefficients for the interaction term are all negative. These coefficients are also all statistically significant, except for those in the export model with only country fixed effects. However, the p-value for FDI inflows and the interaction term are 0.097 and 0.068 respectively, only slightly above the significance cutoff of 0.05. Controlling variables tell a similar story to that of the

Table 3*Model Group With Interaction Terms*

Variable	Export Model 1	Export Model 2	Import Model 1	Import Model 2
FDI Inflows	2.4810. (1.4850)	3.4070* (1.5230)	3.2840* (1.6210)	4.1360* (1.6070)
FDI Inflows $\times$ GDP	$-4.63e^{-6}$. (2.52 e^{-6})	$-6.23e^{-6}$ * (2.57 e^{-6})	$-5.59e^{-6}$ * (2.74 e^{-6})	$-7.01e^{-6}$ * (2.71 e^{-6})
Country Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	No	Yes	No	Yes
R ²	0.9680	0.9731	0.9561	0.9645
Within R ²	0.6246	0.4521	0.5384	0.3269
Observations	136	136	136	136

Standard errors in parentheses. Full regression results are reported in Appendix Table A2.

Significance codes: . p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

first batch of models, repeating the same pattern in terms of coefficients and statistical significance. R-squared values remain similar to the first batch of models at around 96% with some very slight improvements, while Within R-squared values see much more defined improvements across all four models. Specifically, Within R-squared goes up by 2-3 percentage points for country fixed effects models, and 6-8 percentage points for both country and year fixed effects models.

The data is validated by checking the correlation matrix of the independent variables, as shown in **Table 4**. Overall, the correlations look good, except for the relationship between GDP and population, and between FDI and the IEF. However, these three variables capture different dimensions of a country: economic productivity, market size, and economic policy openness. Furthermore, omitting these may create omitted variable bias.

The normality of the residuals of the four models is validated using Q-Q plots, which is shown in **Figure 1**. Overall, the residuals mostly follow the normal distribution, with deviations in the tail ends. This is normal for real-life economic data, and the model also utilizes robust standard errors (from the `fixest` package in R), so these deviations should not affect the integrity of the model.

Table 4
Correlation Matrix

	Inward FDI	GDP	Unemployment	Population	Industry	Economic Freedom
Inward FDI	1.0000	0.2847	0.2062	-0.0636	-0.4002	0.7004
GDP	0.2847	1.0000	0.1946	0.8596	-0.0202	0.2770
Unemployment	0.2062	0.1946	1.0000	0.1005	0.5165	0.4339
Population	-0.0636	0.8596	0.1005	1.0000	0.0785	-0.1637
Industry	-0.4002	-0.0202	0.5165	0.0785	1.0000	-0.1075
Economic Freedom	0.7004	0.2770	0.4339	-0.1637	-0.1075	1.0000

6. Discussion

6.1. Result Interpretations

The coefficients control variables - GDP, unemployment rate, population, industry and economic freedom - while not within the scope of this study, are worth mentioning. First, GDP consistently has a positive and significant effect on trade, which is in line with existing economic theory. What is surprising is that unemployment rate has a statistically and economically significant positive effect on trade value. This means that the more unemployed the population is, the higher the trade activity, which is somewhat counterintuitive. Third, population has a negative correlation to trade value, which is also somewhat counterintuitive. Fourth, it is expected that the more industrialized a country is, the more trade will occur, mostly thanks to the manufacturing sector. So, the fact that the industry level has a positive effect is unsurprising. Finally, the positive effect of economic freedom on trade is also understandable: more open trade and free trade agreements that facilitate more trade leads to a higher economic freedom score. To sum up, the effects of these controlling variables, while some results are expected, other results deserve further investigation.

The statistical insignificance of the FDI coefficients across all four models in the first batch means that there is no strong evidence of FDI inflows affecting trade value. Furthermore, with one coefficient being negative, even if the results for this first batch of model were statistically significant, the economic implications would be ambiguous. The results from these baseline models suggest that there is little evidence that FDI uniformly

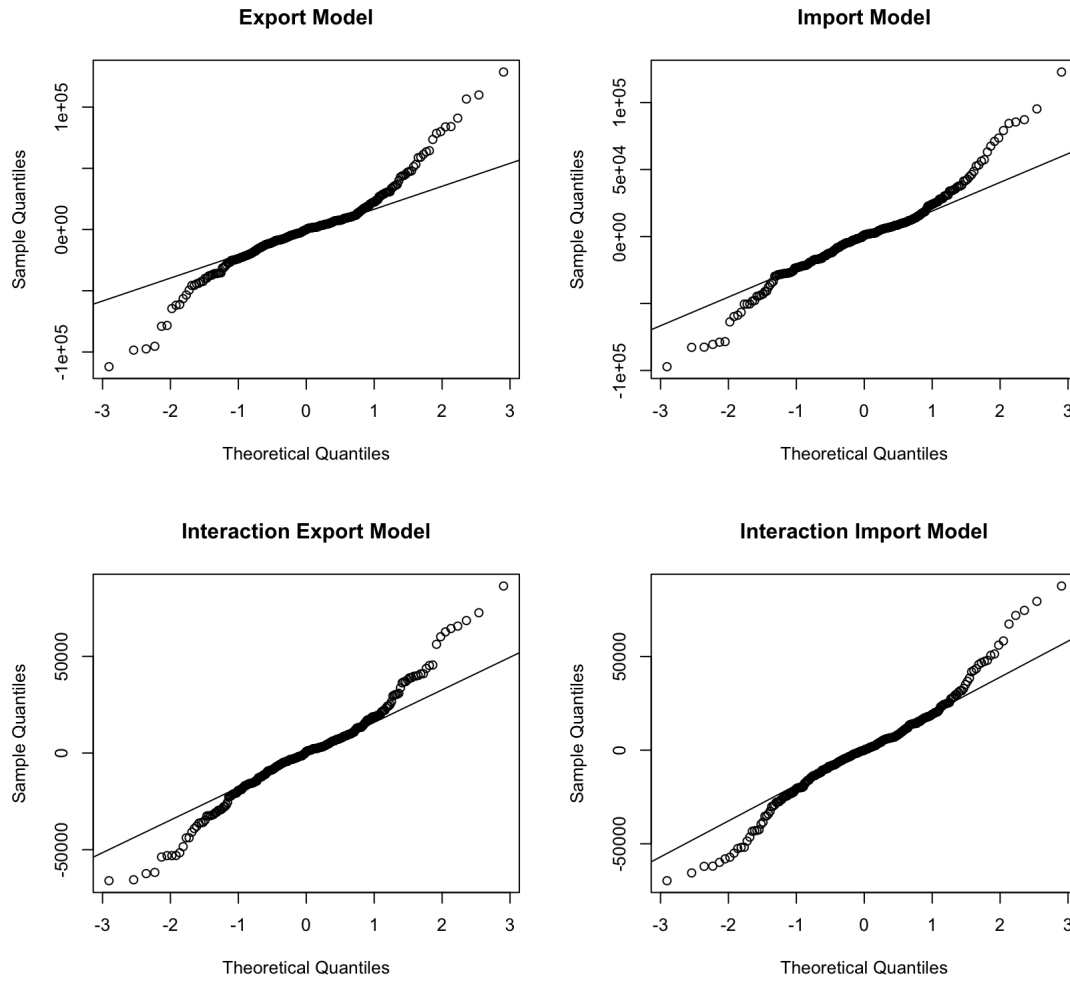


Figure 1
Q-Q plots of residuals of the fixed effects models. Created using R.

affects ASEAN economies. This result makes sense because **ASEAN** contains very heterogeneous economies, with a mix of middle- and high-income countries (Intal et al., 2017). As such, it is reasonable to assume that FDI affects countries differently based on their development level, which GDP is a suitable proxy.

This assumption is supported by the results of the second batch of models, which introduces GDP as an interaction term with FDI inflows. The coefficients for the FDI inflows alone in these models are irrelevant by themselves, as they represent the change of trade value at \$0 GDP. What is interesting is the negative coefficients of the interaction terms across all four models. They represent the fact that when considering the marginal

effect of FDI inflows on trade, lower GDP countries see a greater increase in trade value, both import and export, as FDI inflows increase. This can be attributed to the effects of technology transfer, new processes, and higher-quality employee training mentioned by Borensztein et al. (1998) and Alfaro (2016), all of which being more impactful for lower GDP countries. Since they have worse starting conditions than higher GDP countries (who already has good infrastructure, high-quality labor, etc.), the marginal effects of FDI is larger. This batch of models also have good explanatory powers, with very high R-squared values and relatively high Within R-squared values. The fact that Within R-squared values are higher when controlling only for country fixed effects is understandable: when additionally capturing year fixed effects, the year variable is omitted, so the models lose some explanatory power.

6.2. Limitations

First, while the fixed effects model controls for unobserved, time-invariant differences across countries, it does not address the possibility of reverse causality or interdependence between trade and FDI. Such relationships have been found in existing research such as Liu et al. (2001) or Ghosh (2007). Therefore, the estimated relationships should be interpreted as associations rather than definitive causal effects. Second, the model is susceptible to omitted variable bias. Due to data availability, only GDP, unemployment rate, population, industry level, and economic freedom are introduced as control variables. Adding more controlling factors may increase the robustness and accuracy of future analysis. Notably, the IEF data for Brunei Darussalam is missing from 2010 to 2013, which leads to 4 observations being dropped. However, considering the size of the data, and the relatively smaller role Brunei plays in the economics of the region, the impact of this missing data should be limited. Third, the time frame for this analysis is relatively short - 14 years, due to data availability. Future research can incorporate longer time periods, as more data become available in the future.

7. Ethical Considerations and Implications

7.1. Ethical Considerations

Regarding data usage, the data used in this paper is collected from three sources: ASEANstats, the World Bank, and the Heritage Foundation. Usage of data from the first two sources is granted through their Terms and Conditions of Use and the Creative Commons license, respectively. They are properly cited and acknowledged throughout this paper, which satisfies their terms of use. As for the Heritage Foundation's IEF data, they do not properly license their data, nor do they mention terms of data usage. However, considering that the data is publicly and freely available, common practices of using publicly available data is practiced, including properly acknowledging that the Heritage Foundation as the data owner.

Besides data usage permissions, the authority of the data sources is also a possible point of ethical contention. First, ASEANstats is the official statistical body of ASEAN, and the data it provides are reported by member governments. While this ensures official representation, it also means that accuracy may vary depending on each country's reporting capacity or transparency standards. Second, the World Bank Group is a collective of five international organizations that makes up the largest development bank in the world, with observer status in the United Nations Development Group. They provide standardized global economic indicators and is widely regarded as a credible, neutral data source. However, they also rely on data reported by the governments, which raise similar issues to ASEANstats. Third, the Heritage Foundation is an American conservative think tank. While they publicly display the methodology of calculating the IEF, it is not fully transparent. Furthermore, its conservative political affiliations may lead to biases in calculating the Index, with higher scores for countries that align with the Foundation's values and vice versa for countries that do not.

Regarding the results, their interpretation is limited by the non-causal nature of the fixed effects models. While the findings show patterns of association between FDI and

trade, they should not be taken as evidence of a direct causal relationship. Overstating this link may encourage misguided policies that emphasize attracting FDI without addressing structural economic factors that influence trade outcomes. Furthermore, since this research uses country-level, sector-agnostic data, it does not capture distributional effects within countries. FDI may disproportionately benefit certain industries, regions, or populations, potentially reinforcing inequality or marginalizing vulnerable groups.

There are also implications for policymakers from this result. First, country policies that encourage or attract FDI should consider their economic development level to ensure that the right type of FDI is attracted. Second, these policies must consider the domestic context, as domestic businesses may be weakened (Zhang, 2001), or the country may lack sufficient human capital (Borensztein et al., 1998). Thus, policies that attract FDI must also be complemented by policies that develop domestic firms and human resources. Third, if a country is looking to bolster its trade amount, it should combine FDI attraction with broader trade policy reforms, rather than relying solely on the former.

8. Conclusion

This paper set out to investigate the relationship between foreign direct investment (FDI) and trade in the ASEAN region - an area marked by sustained economic growth, growing trade volumes, and rising investor interest. This study employed fixed effects regression models to estimate both the direct and conditional effects of FDI on trade value. While the baseline models found no significant direct effect of FDI on trade across the region, the inclusion of an interaction term between FDI and GDP revealed that lower GDP economies benefit more from FDI. This has important consequences for policymakers: the study highlights the need to tailor investment policies not only to attract FDI, but also to ensure that host economies have the infrastructure, labor force, and institutions necessary to translate FDI inflows into meaningful trade gains.

References

- Africano, A. P., Magalhães, M., et al. (2005). FDI and Trade in Portugal: a gravity analysis. *Research Work in Progress*, 174, 1–24.
- Aizenman, J., & Noy, I. (2006). FDI and trade—Two-way linkages? *The Quarterly Review of Economics and Finance*, 46(3), 317–337. Retrieved from <https://www.sciencedirect.com/science/article/pii/S1062976906000305> (Real and Financial Aspects of Financial Integration) doi: <https://doi.org/10.1016/j.qref.2006.02.004>
- Alfaro, L. (2016, 03). Gains from Foreign Direct Investment: Macro and Micro Approaches. *The World Bank Economic Review*, 30(Supplement 1), S2–S15. Retrieved from <https://doi.org/10.1093/wber/lhw007> doi: 10.1093/wber/lhw007
- ASEANstats. (n.d.). *Terms and Conditions of Use*. Retrieved from <https://www.aseanstats.org/terms-and-conditions-of-use/> (Accessed: 2025-02-23)
- Barry, F., & Bradley, J. (1997). FDI and trade: the Irish host-country experience. *The Economic Journal*, 107(445), 1798–1811.
- Borensztein, E., De Gregorio, J., & Lee, J.-W. (1998). How does foreign direct investment affect economic growth? *Journal of International Economics*, 45(1), 115–135. Retrieved from <https://www.sciencedirect.com/science/article/pii/S0022199697000330> doi: [https://doi.org/10.1016/S0022-1996\(97\)00033-0](https://doi.org/10.1016/S0022-1996(97)00033-0)
- Dunning, J. H. (2001). The Eclectic (OLI) Paradigm of International Production: Past, Present and Future. *International Journal of the Economics of Business*, 8(2), 173–190. Retrieved from <https://doi.org/10.1080/13571510110051441> doi: 10.1080/13571510110051441
- Faeth, I. (2009). DETERMINANTS OF FOREIGN DIRECT INVESTMENT – A TALE OF NINE THEORETICAL MODELS. *Journal of Economic Surveys*, 23(1), 165–196.

- Retrieved from <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1467-6419.2008.00560.x> doi: <https://doi.org/10.1111/j.1467-6419.2008.00560.x>
- Feridun, M., & Sissoko, Y. (2011). Impact of FDI on Economic Development: A Causality Analysis for Singapore, 1976-2002. *International Journal of Economic Sciences & Applied Research*, 4(1).
- Ghosh, I. (2007). The relation between trade and fdi in developing countries – a panel data approach. *Global Economy Journal*, 7(3). Retrieved 2025-04-07, from <https://doi.org/10.2202/1524-5861.1272> doi: doi:10.2202/1524-5861.1272
- Intal, J. P., Chen, L., et al. (2017). *ASEAN and member states: transformation and integration*. Economic Research Institute for ASEAN and East Asia.
- Li, Q., & Maani, S. (2018, Jan 01). Detecting positive effects of the ASEAN-China free trade agreement on foreign direct investment. *International Economics and Economic Policy*, 15(1), 69-87. Retrieved from <https://doi.org/10.1007/s10368-016-0366-y> doi: 10.1007/s10368-016-0366-y
- Li, Q., Scollay, R., & Maani, S. (2016). Effects on China and ASEAN of the ASEAN-China FTA: The FDI perspective. *Journal of Asian Economics*, 44, 1-19. Retrieved from <https://www.sciencedirect.com/science/article/pii/S1049007816300288> doi: <https://doi.org/10.1016/j.asieco.2016.05.001>
- Liu, X., Wang, C., & Wei, Y. (2001). Causal links between foreign direct investment and trade in China. *China Economic Review*, 12(2), 190-202. Retrieved from <https://www.sciencedirect.com/science/article/pii/S1043951X01000505> doi: [https://doi.org/10.1016/S1043-951X\(01\)00050-5](https://doi.org/10.1016/S1043-951X(01)00050-5)
- Magasházi, A. (2015). The integraton by trade and FDI of emerging economies: The ASEAN example. *Society and Economy*, 37(2), 207–223.
- Okabe, M., et al. (2015). Impact of free trade agreements on trade in East Asia. *ERIA Discussion Paper Series*, 1, 48.

- Pangarkar, N., & Lim, H. (2003). Performance of foreign direct investment from Singapore. *International Business Review*, 12(5), 601–624.
- The ASEAN Secretariat & United Nations Trade and Development. (2024). *ASEAN Investment Report 2024 - ASEAN Economic Community 2025 and Foreign Direct Investment*. ASEAN Secretariat.
- Wadhwa, K., & Reddy, S. S. (2011). Foreign direct investment into developing Asian countries: The role of market seeking, resource seeking and efficiency seeking factors. *International Journal of Business and management*, 6(11), 219.
- World Bank Group. (n.d.). *Data Access And Licensing*. Retrieved from <https://datacatalog.worldbank.org/public-licenses> (Accessed: 2025-02-23)
- Zhang, K. H. (2001). How does foreign direct investment affect economic growth in China? *Economics of Transition*, 9(3), 679-693. Retrieved from <https://onlinelibrary.wiley.com/doi/abs/10.1111/1468-0351.00095> doi: <https://doi.org/10.1111/1468-0351.00095>

Appendix A

Supplementary Tables

Table A1*Baseline Model Group (Full)*

Variable	Export Model 1	Export Model 2	Import Model 1	Import Model 2
inward_fdi	-0.0118 (0.2998)	-0.0540 (0.2511)	0.2781 (0.3431)	0.2428 (0.2826)
gdp	0.6070*** (0.1040)	0.5226*** (0.1083)	0.4393*** (0.1104)	0.3096** (0.1097)
unemployment_rate	10,099.8** (3,292.3)	8,753.5* (3,622.0)	6,409.3* (2,775.2)	5,460.9 (3,499.2)
population	-0.0041*** (0.0011)	-0.0043*** (0.0011)	-0.0020 (0.0013)	-0.0020 (0.0012)
industry	3,140.8*** (788.9)	1,494.8 (955.7)	2,894.8*** (742.4)	667.0 (924.4)
econ_freedom	222.6 (941.7)	1,784.8 (1,282.2)	172.8 (924.3)	1,872.1 (1,204.0)
Country Fixed Effect	Yes	Yes	Yes	Yes
Year Fixed Effect	No	Yes	No	Yes
R ²	0.9661	0.9701	0.9526	0.9599
Within R ²	0.6017	0.3929	0.5012	0.2398
Observations	136	136	136	136

Significance codes: . p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table A2*Model Group With Interaction Terms (Full)*

Variable	Export Model 1	Export Model 2	Import Model 1	Import Model 2
inward_fdi	2.481. (1.485)	3.407* (1.523)	3.284* (1.621)	4.136* (1.607)
gdp	0.7343*** (0.1290)	0.6981*** (0.1366)	0.5928*** (0.1293)	0.5070*** (0.1336)
unemployment_rate	9,412.7** (2,932.6)	8,747.5** (3,262.4)	5,580.8* (2,345.7)	5,454.1. (3,020.9)
population	-0.0047*** (0.0012)	-0.0046*** (0.0011)	-0.0027* (0.0013)	-0.0024* (0.0012)
industry	2,920.6*** (769.8)	1,492.0 (911.1)	2,629.2*** (731.7)	663.8 (859.5)
econ_freedom	-43.72 (855.9)	1,893.6. (1,080.6)	-148.3 (798.9)	1,994.5* (977.6)
inward_fdi $\times$ gdp	$-4.63e^{-6}$. ($2.52e^{-6}$)	$-6.23e^{-6}$ * ($2.57e^{-6}$)	$-5.59e^{-6}$ * ($2.74e^{-6}$)	$-7.01e^{-6}$ * ($2.71e^{-6}$)
Country Fixed Effect	Yes	Yes	Yes	Yes
Year Fixed Effect	No	Yes	No	Yes
R ²	0.9680	0.9731	0.9561	0.9645
Within R ²	0.6246	0.4521	0.5384	0.3269
Observations	136	136	136	136

Significance codes: . p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Appendix B

GitHub Repository

The data, code, and documentation for this study can be found here:

<https://github.com/mb-linh/da-401>